

Q1 Start Your Cluster

```
(base) on@its-nakul:~$ cd ~/kube
(base) on@its-nakul:~/kube$ minikube start
W0910 14:32:30.988933 5631 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/on/.docker/contexts/meta/37a8ee1ce196
87d132fe29051dcae29d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 14:32:30.989963 5631 main.go:295] Try running 'docker context use default' to resolve the above error
🐳 minikube v1.38.1 on Ubuntu 24.04
💡 minikube 1.39.0 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.39.0
To disable this notice, run: 'minikube config set WantUpdateNotification false'

🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.50 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
👉 Done! kubectll is now configured to use "minikube" cluster and "default" namespace by default
(base) on@its-nakul:~/kube$ kubectl get node
NAME                STATUS    ROLES    AGE   VERSION
minikube            Ready     control-plane  23d   v1.35.1
(base) on@its-nakul:~/kube$
```

Q2 Run Your First Pod

```
on@its-nakul:~/kube$ minikube start
W0910 14:43:56.995899 16665 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/on/.docker/contexts/meta/37a8ee1ce196
87d132fe29051dcae29d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 14:43:56.995948 16665 main.go:295] Try running 'docker context use default' to resolve the above error
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👉 Done! kubectll is now configured to use "minikube" cluster and "default" namespace by default
(base) on@its-nakul:~/kube$ kubectl get pods
No resources found in default namespace.
(base) on@its-nakul:~/kube$ kubectl run myweb --image=nginx
pod/myweb created
(base) on@its-nakul:~/kube$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
myweb       1/1     Running   0           7s
(base) on@its-nakul:~/kube$
```

Q3 Delete a Pod and See What Happens

```
(base) on@its-nakul:~/kube$ minikube start
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87d132fe29051dcae29d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
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🐳 minikube v1.38.1 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
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📡 Pulling base image v0.0.50 ...
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   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
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👉 Done! kubectll is now configured to use "minikube" cluster and "default" namespace by default
(base) on@its-nakul:~/kube$ kubectl get pods
No resources found in default namespace.
(base) on@its-nakul:~/kube$ kubectl run myweb --image=nginx
pod/myweb created
(base) on@its-nakul:~/kube$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
myweb       1/1     Running   0           7s
(base) on@its-nakul:~/kube$ kubectl delete myweb pods
error: the server doesn't have a resource type "myweb"
(base) on@its-nakul:~/kube$ kubectl delete pod myweb
pod "myweb" deleted from default namespace
(base) on@its-nakul:~/kube$ kubectl get pods
No resources found in default namespace.
(base) on@its-nakul:~/kube$
```

Q4 Create a Deployment and Scale It

```
(base) on@its-nakul:~/k8s$ kubectl create deployment web --image=nginx --replicas=2
deployment.apps/web created
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-6gvd6               1/1     Running   0           9s
web-68d995574f-htnn8               1/1     Running   0           9s
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-6gvd6               1/1     Running   0           13m
web-68d995574f-htnn8               1/1     Running   0           13m
(base) on@its-nakul:~/k8s$ kubectl delete pod web-68d995574f-6gvd6
pod "web-68d995574f-6gvd6" deleted from default namespace
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-d4rdn               0/1     ContainerCreating   0           3s
web-68d995574f-htnn8               1/1     Running             0           13m
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-d4rdn               1/1     Running             0           37s
web-68d995574f-htnn8               1/1     Running             0           14m
(base) on@its-nakul:~/k8s$ kubectl scale deployment web --replicas=5
deployment.apps/web scaled
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-2kfhr               0/1     ContainerCreating   0           14s
web-68d995574f-d4rdn               1/1     Running             0           3m16s
web-68d995574f-htnn8               1/1     Running             0           17m
web-68d995574f-x2ghj               1/1     Running             0           14s
web-68d995574f-xr2jm               1/1     Running             0           14s
(base) on@its-nakul:~/k8s$ kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
web     5/5     5             5           17m
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-2kfhr               1/1     Running   0           35s
web-68d995574f-d4rdn               1/1     Running   0           3m37s
web-68d995574f-htnn8               1/1     Running   0           17m
web-68d995574f-x2ghj               1/1     Running   0           35s
web-68d995574f-xr2jm               1/1     Running   0           35s
(base) on@its-nakul:~/k8s$
```

Q5 Reach It With a Service

```
web-68d995574f-x2ghj               1/1     Running   0           35s
web-68d995574f-xr2jm               1/1     Running   0           35s
(base) on@its-nakul:~/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-68d995574f-2kfhr               1/1     Running   0           4m14s
web-68d995574f-d4rdn               1/1     Running   0           7m16s
web-68d995574f-htnn8               1/1     Running   0           21m
web-68d995574f-x2ghj               1/1     Running   0           4m14s
web-68d995574f-xr2jm               1/1     Running   0           4m14s
(base) on@its-nakul:~/k8s$ kubectl expose deployment web --port=80 --type=NodePort
Error from server (AlreadyExists): services "web" already exists
(base) on@its-nakul:~/k8s$ kubectl expose deployment web --port=81 --type=NodePort
Error from server (AlreadyExists): services "web" already exists
(base) on@its-nakul:~/k8s$ kubectl expose deployment name --port=80 --type=NodePort
Error from server (NotFound): deployments.apps "name" not found
(base) on@its-nakul:~/k8s$ kubectl expose deployment web --port=81 --type=NodePort
Error from server (AlreadyExists): services "web" already exists
(base) on@its-nakul:~/k8s$ kubectl expose deployment myweb --port=81 --type=NodePort
Error from server (NotFound): deployments.apps "myweb" not found
(base) on@its-nakul:~/k8s$ kubectl expose deployment web-68d995574f-2kfhr --port=80 --type=NodePort
Error from server (NotFound): deployments.apps "web-68d995574f-2kfhr" not found
(base) on@its-nakul:~/k8s$ kubectl expose deployment web-68d995574f-2kfhr
Error from server (NotFound): deployments.apps "web-68d995574f-2kfhr" not found
(base) on@its-nakul:~/k8s$ kubectl expose deployment web-68d995574f-2kfhr --port=80 --type=NodePort
Error from server (NotFound): deployments.apps "web-68d995574f-2kfhr" not found
(base) on@its-nakul:~/k8s$ kubectl expose deployment web-68d995574f-2kfhr
Error from server (NotFound): deployments.apps "web-68d995574f-2kfhr" not found
(base) on@its-nakul:~/k8s$ kubectl expose deployment web --port=80
--port=80 --type=NodePort
Error from server (AlreadyExists): services "web" already exists
(base) on@its-nakul:~/k8s$ kubectl get svc
error: the server doesn't have a resource type "svg"
(base) on@its-nakul:~/k8s$ kubectl get services
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes                           ClusterIP   10.96.0.1     <none>        443/TCP    23d
my-db-postgresql                     ClusterIP   10.99.238.141 <none>        5432/TCP   4d2h
my-db-postgresql-hl                  ClusterIP   None          <none>        5432/TCP   4d2h
test-app                             NodePort    10.106.235.250 <none>        80:31482/TCP 4d3h
web                                  NodePort    10.108.175.98 <none>        80:38947/TCP 18d
```